100% VERIFIED LIVE SCREEN PRICING — ZERO ESTIMATES

Barebone DIY CAN-FD Testbed — Master Bill of Materials

Phase 4 Hardware-in-the-Loop Testbed (Ultra-Low Cost DIY Route)

Post-Quantum Cryptography (PQC) & 3-Tier IDS for In-Vehicle Security Gateway | 2-Bus Physical Architecture

Doc Version: 2.1 (DIY Assembly)
Date: September 17, 2026
Team: Tapan (Lead), Kaustuv, Kunal
Architecture: Pi 5 Gateway + 3 Laptops

TOTAL DIY HARDWARE OUTLAY
₹10,139 / ₹12,718
₹10,139 with lab spare SD | ₹12,718 with new A2 SD

BUDGET SAVED VS. ORIGINAL PLAN
₹27,047 Saved
68% to 74% total expenditure reduction! (~$325 USD)

WAVESHARE ISOLATED HAT
ELIMINATED
Replaced by 2× DIY USB dongles on Pi (saved ₹6,269)

COMMERCIAL CANABLES
ELIMINATED
Replaced by STM32G431 + TJA1051 (saved ₹20,000+)

#	Component & Specification	Vendor / Direct Source Link	Unit Price	Qty	Total Outlay	Engineering & Implementation Role
1	**Raspberry Pi 5 Model B (1GB/2GB RAM)** BCM2712 Quad-Core Cortex-A76 @ 2.4 GHz	Silverline Electronics	₹5,251.00	1	₹5,251.00	CISG Gateway host. 1GB/2GB easily runs ~300 MB CISG + ONNX model. Identical 2.4 GHz compute.
2	**Official 27W USB-C PD Power Supply** 5.1V / 5.0A custom PD profile (White)	Amazon India	₹1,499.00	1	₹1,499.00	Unlocks full 1.6A downstream USB current on Pi 5. Prevents brownout resets under dual adapter load.
3	**Official Pi 5 Active Cooler (SC1148)** Aluminum heatsink + PWM blower fan	Amazon India	₹699.00	1	₹699.00	Clamps SoC <65°C under 5 Mbps Red Team flood load. Guarantees 0.82 ms deterministic ONNX inference.
4	**WeAct STM32G431CBT6 Core Board** 170 MHz Cortex-M4, USB-C, 128KB ROM, native FDCAN	Robu.in (WeAct)	₹559.00	4	₹2,236.00	MCU host for 4 DIY adapters (2 for Pi, 2 for Laptops). Built-in USB DFU flasher. Runs `candleLight`.
5	**NXP TJA1051T/3/1J CAN-FD Transceiver** 5 Mbps data rate, 3.3V logic (VIO), SOIC-8	Robu.in (NXP)	₹76.00	4	₹304.00	High-speed automotive CAN-FD transceiver. Dedicated 3.3V logic supply (pin 5) protects STM32 GPIOs.
6	**SOP-8 to DIP-8 Breakout PCBs** 5-pack adapter boards with 2.54mm pin headers	Robu.in / Local Shop	₹50.00	1 pk	₹50.00	Converts surface-mount SOIC-8 TJA1051 chips to breadboard-friendly pins for easy DuPont jumper wiring.
7	**SanDisk Extreme 64GB MicroSD (A2/V30)** ≥4000 Read IOPS (or spare lab card ≥16GB)	Amazon India	₹2,579.00	1	₹2,579.00 (₹0.00 spare)	Strictly the Pi OS boot drive. Experimental dataset fixtures stay on laptop SSD. Can use lab spare card.
8	**Prakti 2.54mm DuPont Jumpers (40-pin)** Female-to-Female jumper wire ribbon	Amazon India	₹90.00	1 pk	₹90.00	Connects STM32 header pins (5V, 3.3V, GND, TX, RX) to TJA1051 transceiver DIP breakout boards.
9	**120Ω 1/4W Metal Film Resistors** Bus end termination (2 resistors per bus)	Lab Stock / Retail	₹2.50	4	₹10.00	ISO 11898-2 physical termination. 2 resistors per bus placed at physical endpoints (60Ω parallel load).
10	**Cat6 Twisted Pair Cable & USB Cables** 2m patch cable + 4× USB-C data cables	Lab Borrowed / Spares	₹0.00	1 set	₹0.00	Slicing 1m Cat6 gives 4 tightly twisted pairs (100Ω impedance). Phone USB cables connect adapters.
11	**3× Team Laptops & 100MHz DSO Scope**	Team / Univ Lab	₹0.00	1 set	₹0.00	Laptops act as vehicle simulator and Red Team attacker. Oscilloscope verifies real differential wave.

#	Component & Specification	Vendor / Direct Source Link	Unit Price	Qty	Total Outlay	Engineering & Implementation Role
	Tapan, Kaustuv, Kunal machines + Engineering DSO					

TJA1051T/3 to STM32G431 Pinout & Wiring Table

TJA1051 Pin	Pin Name	Connects To	Voltage	Function
1	**TXD**	STM32 Pin **PB9**	3.3V Digital	Transmit data from MCU to transceiver
2	**GND**	STM32 Pin **GND** + CAN_GND	0.0V	Common signal and circuit ground
3	**VCC**	STM32 Pin **5V** (USB rail)	5.0V DC	Powers the high-energy differential bus driver
4	**RXD**	STM32 Pin **PB8**	3.3V Digital	Receive data from transceiver to MCU
5	**VIO**	STM32 Pin **3.3V**	3.3V DC	**CRITICAL:** Sets 3.3V logic levels to protect STM32!
6	**CAN_L**	Physical Bus **CAN_L**	1.5V / 2.5V	Negative differential signal line
7	**CAN_H**	Physical Bus **CAN_H**	2.5V / 3.5V	Positive differential signal line
8	**S**	Tie to **GND**	0.0V	Disables silent mode; enables bidirectional CAN-FD

Open-Source Firmware Flashing & Deployment Guide

1. **Zero Programmer Cost:** The STM32G431 chip includes a factory ROM USB DFU bootloader. No ST-Link debugger hardware required.
2. **DFU Mode Entry:** Hold down the `BOOT0` tactile button on the WeAct board, plug into laptop USB-C, then release `BOOT0`.
3. **Flash Firmware:** Open **STM32CubeProgrammer**, select USB mode, load precompiled `weact_g431_candlelight.bin`, and click **Start Programming** at address `0x08000000`.
4. **Linux Auto-Enumeration:** Plug into Raspberry Pi 5 USB ports. Mainline Linux kernel `gs_usb` driver instantly binds them as `can0` and `can1`.
5. **Windows Laptops:** Run **Zadig** (free, 30-second setup), select device, install **WinUSB** driver. Native Python `python-can` communicates via `interface='gs_usb'`.

Critical Engineering Safeguard: The NXP **TJA1051T/3** transceiver requires two separate power connections: Pin 3 (VCC) must be connected to 5.0V to power the high-speed differential CAN-FD driver, while Pin 5 (VIO) must be connected strictly to 3.3V to clamp RXD logic pulses to safe microcontroller levels. Pin 8 (S) must be tied to GND to disable Silent Mode.